

THIS IS TECHNOLOGY, CULTURE AND SCIENCE WITH A DIFFERENCE, AND THAT DIFFERENCE IS USUALLY ENOUGH TO MAKE MOST NORMAL PEOPLE GO "WTF?"

This month in ALT:

This month in Alt, we delve into Android Auto and Tesla Autopilot, we learn about scitech institutions in India, we speak to Nothing about their first product, and more.



Strange monsoon critters

Check out this fascinating gallery of snakes, frogs, spiders, worms, beetles, crabs, snakes, turtles, birds and lizards that can be spotted during the monsoon season in India.

<https://dgit.in/raincritters>

WHAT'S NEW

Cell phone data use negatively affects Wi-Fi networks

If there are two competing networks to deliver connectivity to devices, each one negatively affects the performance of the other. Since anyone can create Wi-Fi hotspots in the unlicensed spectrum, a number of hotspots in the same area are also in competition with each other. Some telecom service providers, reaching the limits of their allocated licensed spectrum, are beginning to use the unlicensed spectrum, using an approach known as licensed assist access to provide connectivity. By accessing multiple networks, researchers found that competition reduced the speed of data transmitted, the



amount of data transmitted, and the signal quality as well. Basically the idea is that if everyone starts

using the shared spectrum irresponsibly, then it just gets jammed for everyone. The LAA networks establish a continuous connection with the device, much like a cellular tower, but a Wi-Fi network is built from the ground up to support connectivity to multiple devices, queuing up transmission tasks.

<https://dgit.in/wifided>

Dark mode and battery life

Using a new tool known as the Per-Frame OLED Power Profiler (PFOP), researchers are analysing the ways OLED screen consume power from the battery. The dark mode in devices introduced by both Apple and Android was expected to extend the battery life of the device, as the screen would use far less power to display more dark pixels. However, researchers found that most consumers use dark mode in



such a way that it does not lead to any significant savings on the battery life. Between 30 to 50 percent brightness, a change in the dark mode only led to a difference of between 3 to 9 percent of reduction in

power use, for modern OLED phones. At higher brightness levels, the energy savings were also larger, with switching from light mode to dark mode at 100 percent brightness leading to savings of between 39 to 47 percent in battery power used. The researchers also found that using lower levels of brightness in light mode could save as much power as using dark mode at higher brightness levels.

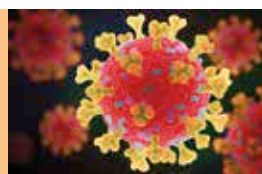
<https://dgit.in/dmbl>



Electricity out of air

An electrical engineer and microbiologist have developed a device that uses a protein and water in the atmosphere to literally pull energy out of thin air. The sustainable Air-Gen uses protein nanowires that are connected to electrodes, does not require sun or wind and works indoors.

<https://dgit.in/airgen>



20,000 year old pandemic

Researchers are looking into the DNA of humans to study how they might have evolved to combat viruses in the past have found the footprint of a coronavirus pandemic that broke out 20,000 years ago, in East Asia. The groundbreaking research helps prevent future outbreaks.

<https://dgit.in/20krona>



Cities make you happy

The rustic rural lifestyles celebrated by cottagecore enthusiasts may be idyllic only in appearance, with researchers finding evidence that the socioeconomic networks and large urban environments found in cities are linked to lower rates of psychological depression.

<https://dgit.in/happycities>